NOTE

Yuqing Rong
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1 Motivation

application: fault tolerant quantum computing.

connection between phase of matter and the application on quantum computer ?

how to find the error from measurement ?

An TO-trivial phase transition occurs at a critical value of the error rate; if the error rate is below the critical value (the accuracy threshold), encoded information can be protected arbitrarily well in the limit of a large code block. find this threshold ?

A measure error (boundary error) comes from multiple possible bulk errors: $P(m) = \sum_E P(m|E)P(E)$

$E_{virtual} -> E_{physical} -> E(m)$

2 Method

2.1 isoTNS representation

We consider the toric code on a square lattice with open boundary conditions,

$$H = -\sum_s A_s - \sum_p B_p, \qquad A_s = \prod_{i \in s} \sigma_i^z, \qquad B_p = \prod_{i \in p} \sigma_i^x.$$

Each bond is represented by a doubled Hilbert space, $|i\rangle \mapsto |ii\rangle$. The local rank-eight tensor is

$$T^{ijkl}_{\alpha\beta\gamma\delta} = \begin{cases} \frac{1}{\sqrt{2}}\,\delta_{i\alpha}\delta_{j\beta}\delta_{k\gamma}\delta_{l\delta}, & \text{if } \alpha+\beta+\gamma+\delta = 0 \pmod 2, \\ 0, & \text{otherwise.} \end{cases}$$

The physical and virtual indices are ordered as east, north, west, and south.

Independent bit-flip errors occur on the internal virtual bonds with probability p. A distance-d patch contains $2d(d-1)$ such bonds.

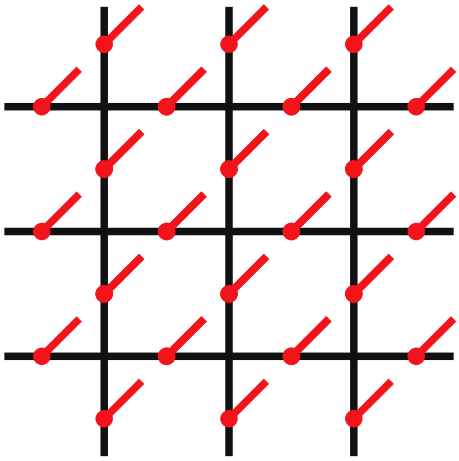


Figure 1: Square-lattice geometry of the planar code with open boundaries.

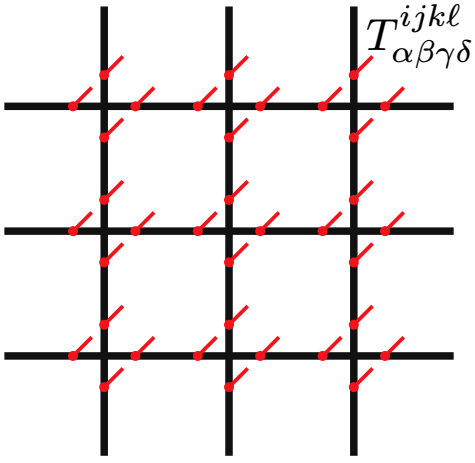


Figure 2: Planar-code lattice in the doubled-Hilbert-space representation. Each virtual bond is doubled as $|i\rangle \mapsto |ii\rangle$, and $T^{ijkl}_{\alpha\beta\gamma\delta}$ labels the local rank-eight tensor.

2.2 Isometric tensors and sequential circuit

2.2.1 Local isometries

The local tensor can be regarded as an isometry and completed to a unitary gate. This correspondence provides a circuit representation of the tensor network.

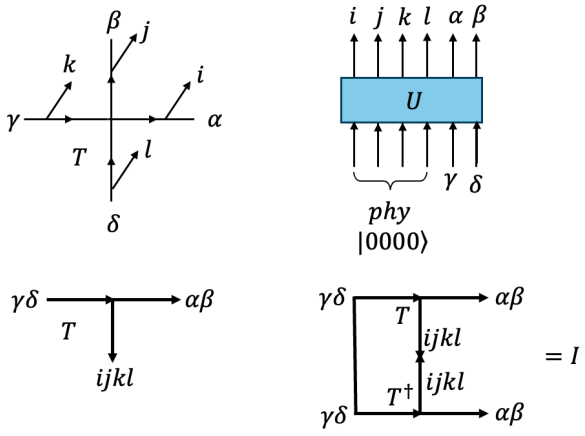


Figure 3: Local tensor T as an isometry and its completion to a unitary gate U. The input legs γ, δ, together with four physical ancillas initialized in $|0000\rangle$, are mapped to the output legs $\alpha, \beta, i, j, k, l$; the lower panels show the reshaped tensor and the isometry condition $T^\dagger T = I$.

2.2.2 Boundary conditions and gate schedule

The planar patch has two rough and two smooth boundaries and encodes one logical qubit. Physical qubits are initialized in $|0\rangle$. The virtual boundary convention is $|+\rangle$ on the west and east boundaries and $|0\rangle$ on the south and north boundaries. At the east and north boundaries, the terminal carrier is absorbed into the local gate.

There are four local-gate types: a six-qubit bulk gate, five-qubit east and north boundary gates, and a four-qubit corner gate. Each gate satisfies the isometry condition. Gates are applied in diagonal layers, progressing from south to north and from west to east.

2.3 Sampling and virtual-bond diagnostics

The circuit admits a classical message-passing description. The Yao.jl implementation is a small-system reference: it applies each local gate to a state vector, performs a Born measurement, records the result, and resets the measured physical qubits. The production estimator instead works directly with batches of virtual-bond error frames. For each shot, it samples independent horizontal and vertical bond-error bits with probability p, then evaluates the detector record and west-boundary logical parity by XOR operations. The full detector batch is decoded once with PyMatching, and a

logical failure is counted when the predicted logical parity differs from the sampled one. No unitary gates or individual physical measurement outcomes are simulated.

2.3.1 Virtual-bond noise checks

For an $L \times L$ lattice, with $L = d$, the following quantities diagnose the virtual-bond noise and its percolation behaviour:

1. *Injected-error and mismatch densities:*
$$\rho_{\text{injected}} = \frac{\text{number of erroneous virtual bonds}}{2L(L-1)+4L} \approx p,$$
$$\rho_{\text{mismatch}} = \frac{\text{number of mismatched internal doubled bonds}}{2L(L-1)} \approx p.$$

2. *Plaquette frustration:* the probability that a plaquette has odd error parity is
$$P_{\text{odd}} = \binom{4}{1} p(1-p)^3 + \binom{4}{3} p^3(1-p) = \frac{1}{2}\left[1-(1-2p)^4\right].$$

3. *Largest occupied cluster:* the normalized cluster size is $S_{\max}/L^2$, where $S_{\max}$ is the number of sites in the largest connected error cluster.

4. *Horizontal spanning probability:*
$$P_{\text{span}}^{\text{horizontal}} = P(\text{an error cluster connects the west and east boundaries}).$$

2.4 Syndrome extraction

A detector is a binary indicator of whether an error violates an expected parity constraint. Let $\mathbf{e}$ denote the binary virtual-bond error vector. The detector record is

$$\mathbf{s} = \mathsf{H}\mathbf{e} \pmod 2,$$

where H is the parity-check matrix. Equivalently,

$$s_i = \begin{cases} 0, & \text{if an even number of adjacent bonds have errors,} \\ 1, & \text{if an odd number of adjacent bonds have errors.} \end{cases}$$

For bit-flip noise, we use two detector families:

1. an elementary plaquette detector,
$$S_{r,c} = h_{r,c} \oplus h_{r+1,c} \oplus v_{r,c} \oplus v_{r,c+1};$$

2. north and south boundary detectors,
$$n_c = h_{1,c}, \qquad s_c = h_{d,c}.$$

The total number of detectors is $(d-1)^2 + 2(d-1) = d^2 - 1$.

2.5 Decoding and logical-failure estimation

PyMatching returns both a minimum-weight correction and the logical parity associated with that correction. Let E be the physical error, C the decoder correction, and $R = E \oplus C$ the residual error. A logical failure occurs when

$$l(R) = l(E) \oplus l(C) = 1.$$

The finite-size scaling ansatz for the logical-failure probability is

$$P_{\text{fail}}(d, p) \simeq F\Big((p - p_c)d^{1/\nu}\Big).$$

For each (d, p) point and each batch, we:

1. sample all $2d(d-1)$ virtual-bond errors for multiple shots;
2. compute the $d^2 - 1$ detector values algebraically;
3. compute the hidden west-boundary logical parity;
4. decode the batch with PyMatching; and
5. count disagreements between the actual and predicted logical parities.

For N shots with F logical failures,

$$P_{\text{fail}} = \frac{F}{N}, \qquad \text{SE}(P_{\text{fail}}) = \sqrt{\frac{P_{\text{fail}}(1 - P_{\text{fail}})}{N}}.$$

2.6 Bootstrap uncertainty

To quantify the uncertainty of a crossing estimate, we partition the shots at each (d, p) point into equal batches. Each bootstrap replicate resamples these batch failure rates with replacement, recomputes the monotone logical-failure curves for adjacent code distances, and extracts their unique crossing. The confidence interval is obtained from the corresponding empirical quantiles of the valid bootstrap crossings.

3 compile with Clifford gate

To encode the $|0_L\rangle$ state, we propose two ways:

1. based on $|0_L\rangle \propto (I + \bar{Z}) \prod_s (I + A_s)|+\rangle^{\otimes N}$

2. based on $|0_L\rangle \propto \prod_p (I + B_p)|0\rangle^{\otimes N}$

Here is an $3 * 3$ example:

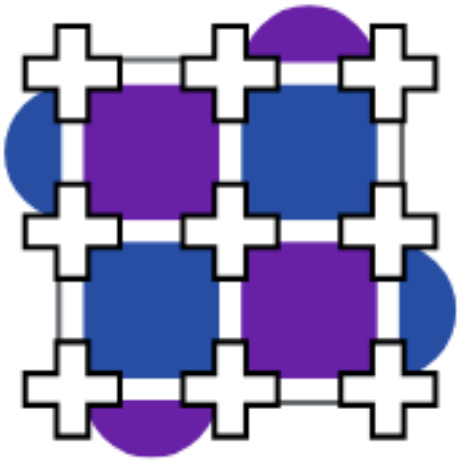


Figure 4: $3 * 3$ lattice. purple: A_s, blue: B_p

1. A_s-based encoding:

(a) A CNOT can establish an even-parity constraint. eg:

$$Z_2 Z_3 Z_5 Z_6 = +1 \quad \Longleftrightarrow \quad x_2 \oplus x_3 \oplus x_5 \oplus x_6 = 0.$$

If qubit 5 starts in $|0\rangle$, the CNOTs

$$2 \to 5, \qquad 3 \to 5, \qquad 6 \to 5$$

compute $x_5 = x_2 \oplus x_3 \oplus x_6$, establishing that parity.

(b) choose qubit 2 to fix the logical-zero parity, 4,5,7,9 as fresh parity targets. These qubits start in $|0\rangle$;

leaves $1, 3, 6, 8$ as independent inputs, these qubits start in $|+\rangle$, so we apply

$$H_1, \ H_3, \ H_6, \ H_8.$$

(c) $\bar{Z} = Z_1 Z_2 Z_3$. Apply

$$1 \to 2, \qquad 3 \to 2.$$

Since x_2 started at zero, it becomes $a \oplus b$. Therefore,

$$x_1 \oplus x_2 \oplus x_3 = a \oplus (a \oplus b) \oplus b = 0.$$

This establishes $\bar{Z} = +1$.

(d) Build the four As constraints block by block.

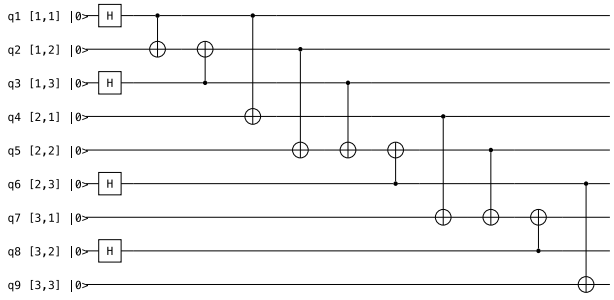


Figure 5: A_s-based encoding for $3 * 3$ lattice.

2. B_p-based encoding:

choose represensitive qubits for every B_p plaquette, and apply H, others are target qubits.

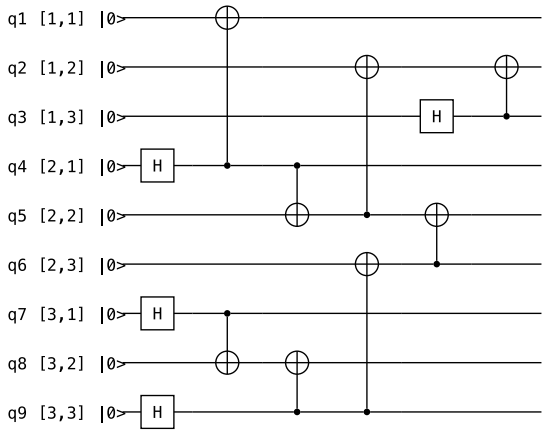


Figure 6: B_p-based encoding for $3 * 3$ lattice.

4 error model

apply X-only or Z-only error after each elementary gate (H, CNOT) with rate p

5 sampling

Monte Carlo sampling chooses which faults occur; the Pauli-frame simulator determines their effect on the circuit.

For each code distance d and physical error probability p:

1. Sample faults. After every gate, independently apply an X fault with probability p to each participating qubit. Each CNOT therefore has two possible fault locations.

2. Propagate errors. Track the X/Z error pattern through the remaining circuit using exact Clifford rules,

$$E \longrightarrow UEU^{\dagger}.$$

This accounts for errors spreading and cancelling, without storing the full wavefunction.

3. Decode and judge success. Calculate the final, ideal stabilizer outcomes and give them to the matching decoder. Compare its proposed correction with the actual propagated error. For preparation of $|0_L\rangle$, a remaining logical X counts as failure.

4. Repeat and estimate. Run N independent trials and count F failures:

$$\boxed{\widehat{P}_L(d,p) = \frac{F}{N}.}$$

Error propagation is exact for the Clifford circuit and Pauli-noise model; the estimated failure probability has finite-sampling uncertainty.

6 results(2D)

1. The two Z-only panels

The Z-only noise at most cause Logical-Z faults

$$\bar{Z}|0_L\rangle = |0_L\rangle.$$

Logical-Z faults can occur without changing the $|0_L\rangle$ state. Maybe need the logical-Z failure curves?

2. The X-only panels

The As/X is fragile logical-parity preparation. Single X fault during logical-state preparation can become a complete logical X error eg, in Figure 4, only X_2 would cause a whole logical-X string.

In the parity part, for m independent locations, their odd-flip probability is

$$q_m(p) = \frac{1-(1-2p)^m}{2},$$

which imposes a logical-failure floor. Increasing m makes this floor larger, approaching 1/2, not smaller.

contain more CNOTs, so more fragile to X-only noise.

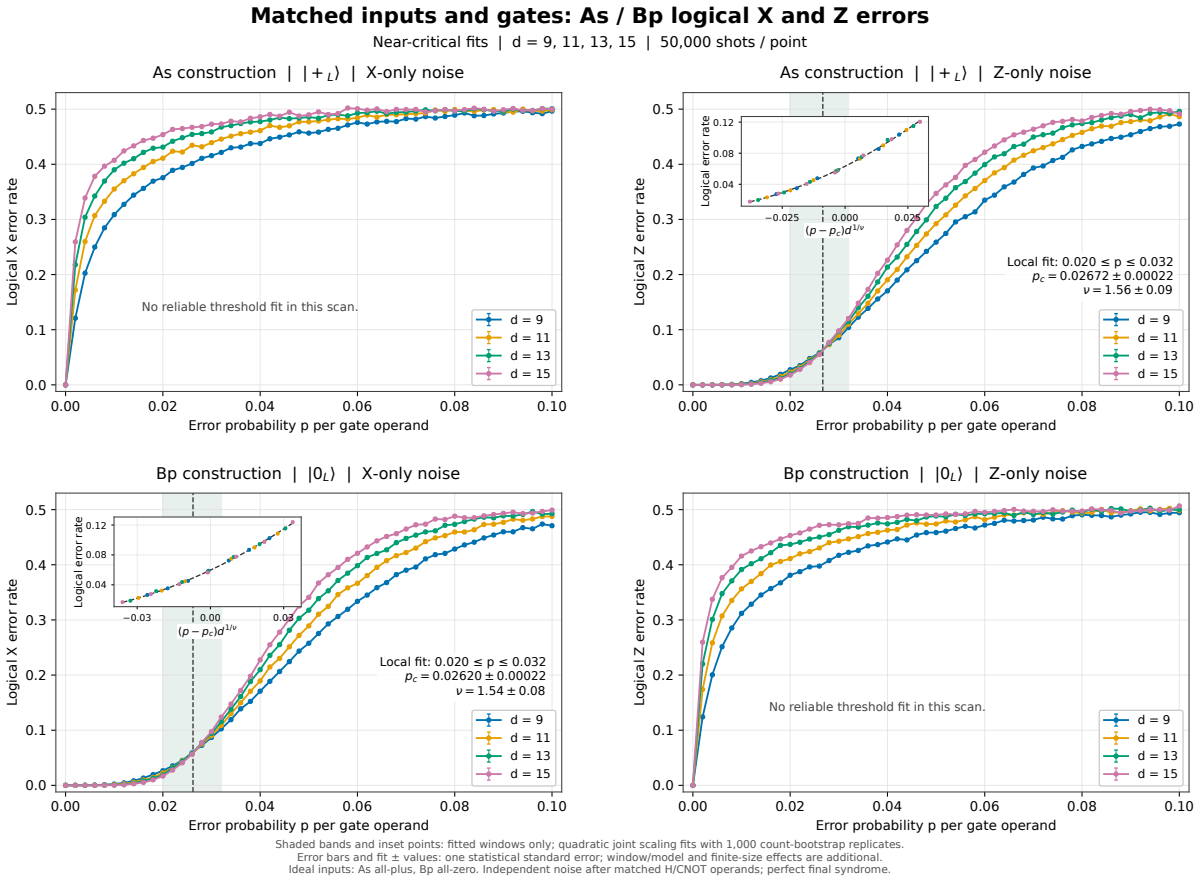


Figure 7: X error and Z error. As based and Bp based circuit is just for different basis, they are symmetry. Bp/X: X propagate only form short-range error; Bp/Z: Z propagate form long-range error.

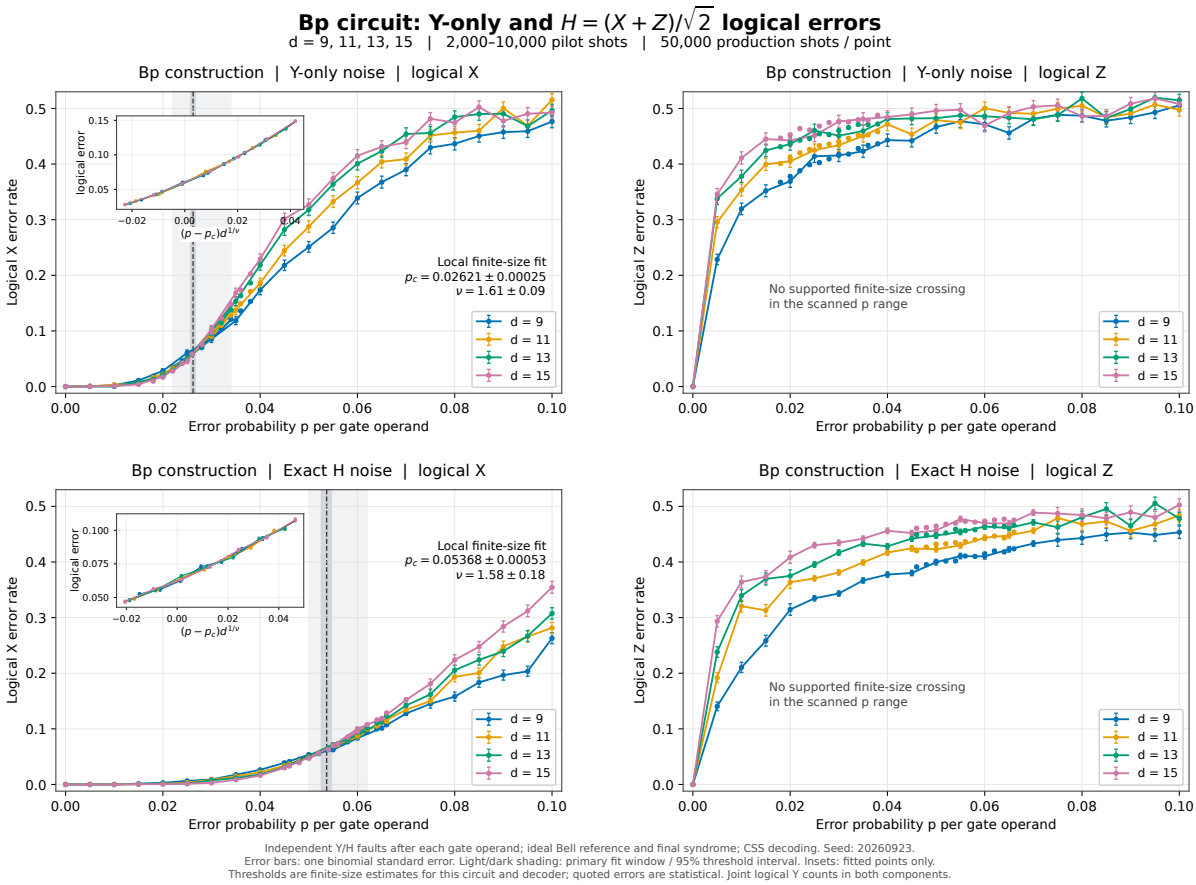


Figure 8: Y error and (X+Z)/$\sqrt{(2)}$ error. Y: seperate decoder treat X and Z separately. (X+Z)/$\sqrt{(2)}$: rotate an angle only change prefactor, the logical Z threshold would always be 0 only if Z element is not zero.

7 3D toric code

7.1 Noise and logical observables

Each active H/CNOT output independently acquires an X or Z fault with probability p in separate single-channel experiments. Initialization, idles, measurements, and corrections are ideal unless stated otherwise. Compared protocols reuse the same fault records. Final recovery uses MWPM for Z and BP+OSD for X; complete recoveries must reproduce the syndrome.

Logical failure is the union of the scored residual logical-Pauli bits. For $|0_L\rangle$ or $|00_L\rangle$, only logical X is preparation failure; logical Z is a Pauli-frame diagnostic. For $|++\rangle$, logical Z is state failure. All threshold estimates below depend on the circuit, decoder, and ideal-feedback assumptions.

7.2 Cartesian sweep: geometry and Z-fault bound

Use a cubic slab with rough open x ends and periodic y, z directions. Retained vertices have $x = 1, \ldots, L_x$; longitudinal edges have $x = 0, \ldots, L_x$. For $L_x \geq 1$, $L_y, L_z \geq 2$, and $M = L_y L_z$,

$$H = -\sum_v A_v - \sum_f B_f, \qquad A_v = \prod_{e \ni v} X_e, \quad B_f = \prod_{e \in \partial f} Z_e,$$

$$n = (3L_x + 1)M, \quad N_A = L_x M, \quad N_B = (3L_x + 2)M, \quad \operatorname{rank} B = (2L_x + 1)M - 1, \quad k = 1.$$

Rough-end plaquettes are triangular, e.g.,

$$B^{xy}_{0,y,z} = Z^x_{0,y,z} Z^x_{0,y+1,z} Z^y_{1,y,z}, \qquad B^{xy}_{L_x,y,z} = Z^x_{L_x,y,z} Z^x_{L_x,y+1,z} Z^y_{L_x,y,z}.$$

Interior plaquettes have four edges; period-two parallel edges remain distinct. The longitudinal Z string and transverse X membrane give

$$d_Z = L_x + 1, \qquad d_X = M, \qquad |0_L\rangle = 2^{N_A/2} \prod_v \frac{I + A_v}{2} |0\rangle^{\otimes n}.$$

Visit vertices in increasing (x, y, z) order. Apply H to the fresh outgoing x-edge, then CNOTs from it to the incoming $x, +y, -y, +z, -z$ edges, in that order. Final-only recovery acts once; layer feedback decodes each completed prefix slab, including its moving triangular frontier, and corrects only available qubits.

Indistinguishable fault pair. For each (y, z), choose Z-fault bits a on the left boundary edge after its first CNOT and b on the outgoing edge after the final CNOT of the first-plane gadget. At checkpoint j, toggling both adds

$$D_j = \prod_{x=0}^{j} Z^x_{x,y,z},$$

which has zero prefix syndrome and becomes logical Z at $j = L_x$. Thus the syndrome history cannot distinguish the pair assignments with the same $a \oplus b$. Even if all other faults and all M pair

XORs are revealed, an odd XOR leaves unbiased logical parity. For an even XOR, the parity bias is $(1-2p)/q$, where $q=(1-p)^2+p^2$. Hence, for $0 \le p < 1/2$ and any syndrome-only decoder,

$$P_{Z,\text{diagnostic}} \ge \frac{1-q^M[(1-2p)/q]^M}{2} = \frac{1-(1-2p)^M}{2}.$$

This bounds the Cartesian logical-Z diagnostic, not $|0_L\rangle$ infidelity or the diagonal circuits below.

With 10000 paired records per point at $L=3,5,7,9$ and $p \in \{1,3,10,30,100,300\}\times 10^{-6}$, the linearly interpolated one-percent feedback diagnostic occurs at

$$\big(p_{1\%}(3), p_{1\%}(5), p_{1\%}(7), p_{1\%}(9)\big) = (1.39\times10^{-4}, 5.03\times10^{-5}, 2.92\times10^{-5}, 1.51\times10^{-5}).$$

The fit $p_{1\%}(L) \propto L^{-1.97}$ has bootstrap exponent interval $[1.77, 2.17]$, consistent with the bound's vanishing diagnostic threshold.

7.3 Diagonal H+5: direct vertex projectors

Set $t=x+y+z$, $u=x$, $v=y$, with open $t=0,\dots,T$ and periodic $(u,v)\in \mathbb{Z}_{L_u}\times\mathbb{Z}_{L_v}$. Edge $E[d,t,r]$ joins $V(t-1,r-s_d)$ to $V(t,r)$, where $s_1=(1,0)$, $s_2=(0,1)$, $s_3=(0,0)$. The slab has

$$n=3(T+1)L_uL_v, \qquad k=2, \qquad d_Z=2\min(L_u,L_v), \quad d_X=(T+1)\min(L_u,L_v).$$

Period-two edges remain distinct. For $V(t-1,r)$, write $e_d=E[d,t-1,r]$ and $n_d=E[d,t,r+s_d]$. The direct star-projector circuit is

$$H(n_1), \quad \mathrm{CX}(n_1,n_2), \quad \mathrm{CX}(n_1,n_3), \quad \mathrm{CX}(n_1,e_1), \quad \mathrm{CX}(n_1,e_2), \quad \mathrm{CX}(n_1,e_3).$$

It has six gates and eleven noisy outputs per cell, implementing a paper-motivated construction rather than a circuit reproduced from it.

Product zero fixes both logical-Z cycles. Ideally project and correct the lower white X triads, run the bulk circuit, then project the upper X triads. Their clean outcomes b are uniform with even parity. If Z faults flip them by u, the physical controller sees only $a=b\oplus u$. For its deterministic correction C, the relative frame is

$$z' = z \oplus C(a) \oplus C(b).$$

Use edge-ID-ordered breadth-first paths to $A_{T-1,0,0}$ after removing both logical-X supports; this guarantees zero logical overlap. The clean branch b and fault record remain hidden from the decoders.

Feedback protocols. Reuse each fault record and boundary branch in four arms. Feedback occurs after $t=1,\dots,T-1$:

- `none`: boundary completion and final recovery only.
- `one_step`: X uses MWPM on the periodic six-edge Q_t checks and applies the first and last path edges (once for a length-one path). Z matches the retiring/frontier star strip, adding one future dummy only for odd parity, and corrects retiring endpoints only. Frontier defects may remain.

- `window_complete`: recover on $E_{t-1} \cup E_t$ using BP+OSD for completed plaquettes and open-boundary MWPM for the star strip. All completed checks clear.
- `full_prefix`: recover from all completed checks on $E_0, \ldots, E_t$; future qubits are unavailable.

The arms change measurements, decoder, residual constraints, and correction support together. Their differences do not isolate correction range. Final BP+OSD uses 50 minimum-sum iterations and `OSD_CS` order two. MWPM minimizes physical length with deterministic ties. Both omit circuit-induced correlations.

Preliminary results. With $T = L_u = L_v = L$, the completed $L = 3, 5, 7$ run contains 60 X-only and 135 Z-only points, 1000 paired records per point, and no invalid recoveries. Table 1 shows selected logical-union rates.

Channel	p	L	None	One step	Window	Prefix
X	0.039448164	3	13.6	14.8	17.7	18.2
X	0.039448164	5	6.5	12.3	17.1	17.1
X	0.039448164	7	4.2	8.6	15.0	15.0
Z	0.0096470592	3	51.6	50.2	41.0	41.1
Z	0.0096470592	5	74.5	75.5	64.7	64.7
Z	0.0096470592	7	76.3	74.4	71.7	71.7

Table 1: H+5 logical-union rates (%). X is $|00_L\rangle$ preparation failure; Z is a Pauli diagnostic. Each row uses 1000 paired records.

Feedback effects depend on p. At $L = 7$, window-complete minus none is +10.8 percentage points at $p = 0.039448164$ (95% interval $[6.99, 14.50]$), but -11.4 points at $p = 0.077446809$ ($[-17.07, -5.62]$). Most Z curves are near the 0.75 union probability of two uniform logical bits, leaving little unsaturated large-size data. These selected contrasts are exploratory; intervals have per-contrast, not across-comparison, coverage.

No H+5 threshold or exponent is estimated. All 14 frozen candidate windows fail the minimum-four-size guard before fitting, so bootstrap fits were not run. This does not imply a zero threshold. Larger sizes remain deferred. Counts, paired contrasts, checkpoints, and audits are documented in `docs/superpowers/toric3d-h5-vertex-projector.md` and its linked `preliminary-l3-7/` results.

7.4 Historical H+6: plaquette transfer

The seven-gate H+6 circuit transfers old controls to fresh target triads on the diagonal slab above, with thirteen noisy outputs per cell. A single Z output fault spreads to at most three qubits in two adjacent slices. Its seed preparation and upper boundary instrument differ from H+5, so their thresholds are not a controlled circuit-only comparison.

For X noise, prepare $|00\rangle$ using ideal seed X-star measurements, with no noisy seed gates. One-step feedback matches defects of the old slice's Z compatibility checks and feeds endpoint repairs forward to fresh targets. Matching is global; this is not a constant-depth spatially local channel.

Final BP+OSD scores the union of logical-X membranes.

For Z noise, prepare $|++\rangle$ using noisy seed H gates followed by ideal Z-compatibility measurements and X corrections. These preserve the seed Z faults. Final MWPM scores the union of logical-Z windings. The older Z scan's intermediate Z-check/X-correction feedback is inactive under pure Z noise.

Separate sector estimates. For $T = L_u = L_v = L$, the Z scan uses $L = 9, 13, 17, 21, 25$ and 10000 shots per point (330000 total). A quadratic logistic fit with a fixed $1/L$ correction gives $p_{c,Z} \simeq 0.0052$. The 0.49%–0.54% range across fit variants measures model sensitivity, not a confidence interval.

The X scan uses $L = 5, 7, 9, 11, 13$, $p = 0.0100$–0.0250, and 1000 paired records per point (35000 total). Final-only curves have no adjacent-size crossings in this window. The all-size fits fail the goodness guard; the accepted cubic sensitivity fit for one-step feedback with $L \geq 7$ gives

$$p_{c,X} = 0.01482, \qquad \nu = 1.056, \qquad p_{\text{deviance}} = 0.01996,$$
$$p_{c,X} \in [0.01428, 0.01543] \quad \text{(conditional 95\% interval)}.$$

Of 1000 bootstrap refits, 999 pass. These separate input-state and channel studies do not establish a joint X/Z threshold or reproduce a threshold from the paper.

Three feedback strengths. A separate paired scan compares no feedback, slice-local feedback, and full-prefix recovery. X slice-local feedback uses the one-step rule above; full-prefix X uses complete BP+OSD. Active Z feedback first corrects seed X stars, then uses either a three-check-sheet band with MWPM on the newest two data slices, or full-prefix MWPM. A single-slice Z window misses old-control faults. Both active Z policies clear all completed stars. Their paths and tie choices can differ. No feedback omits the seed-time correction, so this comparison also changes more than recovery support. Local and prefix check participation scale as $O(TL_uL_v)$ and $O(T^2L_uL_v)$, respectively.

The completed data contain 36000 paired X records at $L = 5, 7, 9, 11$ and 56000 paired Z records at $L = 9, 13, 17, 21$, with no invalid recoveries. No primary fit passes every threshold guard:

- X full-prefix: candidate $p_c = 5.297\%$, rejected by goodness-of-fit $p = 0.00530 < 0.01$ despite adjacent-size crossings. No-feedback and slice-local fits are inconclusive in this window.
- Z slice-local/full-prefix: candidates 3.934%/3.923%, both with missing adjacent-size crossings, nonmonotone fits, and goodness-of-fit below 1.5×10^{-5}.

Nevertheless, at $L = 21, p = 0.035$, Z state-failure rates are 0.7545/0.2445/0.2415 for none/local/prefix. The paired differences are -0.513 for prefix minus none (95% interval $[-0.549, -0.475]$) and -0.003 for prefix minus local ($[-0.0146, 0.0086]$). Thus active Z feedback helps at this point, while a local/prefix threshold difference remains unresolved. The diagnostic candidates do not supersede the separate sector estimates, and failed crossing guards do not imply zero thresholds.

Reports in `docs/superpowers/`: `toric3d-z-threshold.md`, `toric3d-x-feedback-threshold.md`, and `toric3d-diagonal-feedback-strength.md`.

8 questions

8.1 present 2D toric-code

Can a different unitary extension reduce the propagation of Z errors? Can additional resources overcome the 2D obstruction?

introduce majority-like correction in 2D:

(a) redesigned unitary schedule (changes plaquette order, pivot selection, CNOT directions, CNOT order.)

(b) repeated stabilizer measurement rounds + classical decoding and feedback (measurement feedback)

(c) redundant encoding of virtual information and correct it coherently

8.2 move to 3D toric-code

Use measurement feedback for 3D toric code

prepare one 2D layer → measure the layer syndrome → decode → feedback → prepare the next layer.

compare:

(a) naive 3D sequential circuit

(b) feedback-stabilized 3D sequential circuit

(c) logical failure rate versus noise and system size